Strategies for Healthy Living and Longevity

NEXUS Method

for the Performance Profile

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1. Introduction

There are moments in life when we stop, even if only for a fleeting instant, and ask ourselves a question as simple as it is uncomfortable: *Am I living in the best way possible for my body?* It's not a matter of years, or goals, or even success in the conventional sense. It's something more intimate, more silent. It has to do with the quality of each day, with the energy we have upon waking, with mental clarity, with the ability to move, to enjoy life, to rest. Ultimately, it has to do with how we navigate the time we've been given.

We live in an era that, paradoxically, seems to have solved almost everything... and yet, it has complicated what is truly essential. Never before have we had so much access to information, food, technology, and quick fixes. And yet, never before has it been so difficult to discern what is truly important for our health. We navigate amidst contradictory advice, fleeting fads, miraculous promises, and unfounded fears. One diet replaces another, one method displaces the previous one, and amidst this constant noise, common sense seems to have vanished.

This book stems precisely from that need to reclaim what is essential. It doesn't aim to offer magic formulas or instant solutions. It's not a manual for perfection, but an invitation to understand. To understand how our body works, what it truly needs, and why, despite knowing so much, we continue to act against our own nature.

Because the reality is that much of what we do daily isn't a conscious decision, but rather a matter of inertia. We eat at certain times not because we're hungry, but because it's "time." We sit for hours because our routine demands it. We sleep worse than we could because we've normalized habits that interfere with rest. And so, day after day, we repeat patterns without questioning them, as if they were an inevitable part of life.

Breaking that inertia isn't easy. It requires stopping, observing, and, in many cases, unlearning. It means questioning deeply ingrained beliefs, some passed down through generations, others adopted for convenience or due to social pressure. It means accepting that what's usual isn't always right, and that what's comfortable isn't always healthy.

Throughout these pages, we will explore a fundamental idea: health doesn't depend on a single action, but rather on the combination of multiple simple factors that, when aligned, generate a profound impact. It's not about extremes, but about balance. It's about understanding when to eat and when not to, when to move and when to rest, when to stimulate the body and when to allow it to recover. It's about, in a way, returning it to the environment for which it was designed.

We will talk about nutrition, yes, but not from an obsessive perspective of counting calories or following rigid rules, but rather from an understanding of the body's natural rhythms. We will talk about movement, not as an obligation, but as an inherent biological need. We will talk about rest, stress, light, time... all those elements that, although they may seem commonplace, have an enormous impact on how we live and how we age.

We will also debunk myths. Many of them accepted without question, repeated until they become unquestionable truths. Myths that, at times, distance us from what truly

benefits us. Because progress isn't always about adding more things, but about eliminating the unnecessary, simplifying, returning to the basics.

This isn't a book to change your life in a day. It's a book to help you start looking at it differently. To understand that small decisions, sustained over time, have enormous power. To remember that our bodies don't need us to force them, but rather to listen to them.

Ultimately, extending life shouldn't be the only goal. What truly matters is extending life with health, energy, and the ability to enjoy every stage. And that doesn't depend on a secret formula, but on something much more accessible: the will to step off autopilot and start living consciously.

Because, ultimately, it's not about doing more... but about doing better.

2. The problems of today's society

We live in an age of unprecedented abundance, a time when food has ceased to be an uncertain commodity and has become a constant, silent, and almost omnipresent commodity. We need only reach out, open a door, press a button, or walk a few meters to find food. Never before in all of human history has eating been so easy. And yet, therein lies one of the great paradoxes of our time.

Abundance, which in another context would have been a celebrated blessing, has transformed into a constant temptation. Food no longer arrives as a reward after effort, but as a permanent stimulus, available anytime, anywhere. We eat without hunger, out of habit, anxiety, boredom, or simple inertia. The body, designed to manage cycles of scarcity and satiety, is now trapped in an uninterrupted stream of intake that prevents it from remembering its natural rhythm.

This abundance is compounded by the gradual disappearance of the effort required to obtain it. There was a time when eating involved movement, ingenuity, and physical exertion. Today, however, energy comes to us with hardly any investment of our own. The chain that linked hunger, action, and reward has been broken. The act of eating has become detached from effort, and with it, an essential part of its meaning has been lost. The reward has ceased to be earned and has become automatic.

Meanwhile, the world of work has undergone a silent but profound transformation. Jobs that once demanded physical exertion have been gradually replaced by tasks that primarily require the mind—and often not even that entirely. Machines have taken over the burden of physical effort, freeing us from carrying loads, commuting, and manual labor. But this liberation, far from being absolute, has come at a hidden cost: the human body, deprived of its natural function, has fallen into a kind of lethargy.

We spend hours sitting motionless in front of screens that capture our attention but require no movement. Our days pass between chairs, keyboards, and meetings, while our bodies remain in the background, almost forgotten. Muscles, designed for action, weaken; metabolism, designed to adapt to exertion, slows down. And so, almost without realizing it, we drift further and further away from our own nature.

Added to all this is an intangible yet profoundly decisive factor: the stress of modern life. This stress no longer stems from the need to survive in a hostile environment, but from a complex web of work responsibilities, financial worries, social demands, and family challenges. It is a constant, sustained stress that rarely finds an outlet.

Before, tension found its release in action: running, hunting, fighting, moving. Today, however, it accumulates silently. It remains in the body, in the form of restlessness, mental fatigue, insomnia. And often, it seeks relief where it is easiest to find: in food, in consumption, in small, immediate pleasures that offer a momentary respite, but not a real solution.

Thus, abundance, lack of effort, sedentary lifestyles, and stress form a complex web that defines our way of life... and, to a large extent, how we become ill. These are not isolated elements, but pieces of the same puzzle that, when combined, create a lifestyle profoundly different from the one for which we were designed.

And yet, amidst all of this, a silent truth remains: our bodies are still the same. They haven't evolved at the pace of technology or society. They still crave movement, still understand hunger as a signal, still need pauses, effort, and balance. Perhaps the key lies not in rejecting the modern world, but in learning to reconcile ourselves with that ancestral part that still dwells within us, reminding us, with infinite patience, how to live in a way that is more in harmony with who we truly are.

2.1. Abuse of the digestive system

The human digestive system, at its deepest core, is an untouched legacy from our most distant ancestors. It hasn't been shaped by modern conveniences or constant abundance, but by an ancient world where uncertainty dictated the rhythm of life. For thousands upon thousands of years, humankind knew neither a full pantry nor a packed refrigerator; instead, it knew the silence of an empty stomach, the strain of effort, and the brief but intense fullness of the food it had obtained.

Our bodies are written in that ancient language. They weren't designed to receive food continuously, mechanically, and without pause, but to adapt to a natural, almost ritualistic cycle: absence, effort, and reward. First, hunger. That state we try to avoid at all costs today was, in reality, the starting point. Not as suffering, but as a signal, as a call. Hunger sharpened the senses, honed the mind, and awakened instinct. It was the engine that drove human beings to move, to explore, to take risks.

It's sad to say, but nowadays we eat in a completely programmed way and have lost the natural hunger signal. Our brain tells us to eat even when we're not really hungry, more out of habit than necessity.

2.2. Lack of exercise

If we compare ourselves to our ancestors, when they felt true hunger, they had no choice but to submit to movement, action, exercise in its purest form: going out to hunt. It wasn't planned training or a gym routine, but a total expression of the body in search of survival. Walking long distances, running, stalking, climbing, throwing... every muscle had a purpose, every heartbeat a meaning. Effort wasn't optional; it was the only path to life.

In modern society this is no longer the case, and we can obtain food with a caloric expenditure close to 0 in most cases.

Moreover, speaking not only of the physical exertion involved in obtaining food, but also of the physical exertion performed in our workplaces, we could say that the

history of humanity, to a large extent, can be read as the story of how we gradually shifted the burden of effort from our own muscles to the increasingly sophisticated hands of machines. With the arrival of the Industrial Revolution, this process accelerated decisively: what previously required entire days of intense physical labor began to be accomplished through gears, steam, electricity, and, later, automation. The countryside, the factory, transportation... all areas of life began to transform under the same logic: to do more with less human effort.

Initially, this change brought obvious relief. Extreme physical exertion was reduced, certain tasks were shortened, and many living conditions improved. However, a less visible but profoundly significant consequence also occurred, albeit quietly: the human body ceased to be essential for many of the tasks that had defined its function for millennia. What once demanded strength, endurance, and constant movement was replaced by levers, motors, and automated systems that performed the work without fatigue.

Over time, this transformation affected not only the most physically demanding jobs but extended to virtually all areas of work. The office replaced the workshop, the screen replaced manual labor, and the click replaced movement. Today, a large part of the population carries out their daily activities sitting down, with minimal energy expenditure. The body remains active in appearance—it breathes, stands upright, responds—but its actual involvement is minimal compared to that of our ancestors.

This change has a direct impact on calorie expenditure. Humans, designed for constant movement, have seen their daily energy expenditure significantly reduced. Where large amounts of energy were once burned simply to survive—hunting, gathering, building, traveling—today that expenditure must be consciously, almost artificially, sought through planned exercise. Movement has ceased to be a necessity and has become a choice.

The result is a progressive imbalance: we ingest more energy than we expend, not necessarily because we eat more than before in absolute terms, but because our body no longer consumes what it used to. External mechanisms have taken over the work, but our metabolism continues to wait for that level of activity that no longer arrives. And so, little by little, energy accumulates without an outlet, resulting in fatigue, weight gain, and a vague feeling of disconnection from our bodies.

Paradoxically, in our quest to make life easier, we have eliminated much of what kept us in balance. Industrialization has given us time, comfort, and efficiency, but it has also distanced us from the natural physical exertion that was an inseparable part of our biology. Today, more than ever, the challenge is not to produce more with less effort, but to remember that, somewhere within us, that effort is still necessary.

2.3. Unhealthy food

And finally, after the physical exertion, came the reward for our ancestors. Food wasn't immediate or guaranteed, but when it arrived, it did so with an intensity we can scarcely imagine today. Eating wasn't a trivial or automatic act; it was an almost sacred moment. The hunted prey was honored, shared, and celebrated. The body, after the effort, received the energy necessary to recover, strengthen itself, and prepare for the next cycle.

Fortunately for them, the hunted prey or the food they enjoyed was just that: food.

The difference between what our ancestors ate and what we find in supermarkets today is not just a matter of variety, but of essence. They fed on what the earth and nature offered in their purest form: wild animals that had lived freely, fruits picked at their peak ripeness, roots, seeds, and plants that had not been modified or processed. Each food had a story tied to the environment, the climate, and the season; nothing was constant, nothing was artificially prolonged. Today, however, we walk down aisles overflowing with products designed to last, to seduce, and to be consumed effortlessly: processed, refined foods, loaded with sugars, additives, and preservatives, created more in laboratories than in fields. What was once a direct relationship with nature has been transformed into an industry-mediated experience, where flavor is intensified, texture is modified, and availability is endless, but in that process, the true nutritional richness and the original bond between human beings and their food are often diluted.

Thus, food was not a continuous line, but a rhythm, a dance between scarcity and abundance. Hunger, movement, and satiety formed a perfect balance, deeply rooted in our biology. Today, however, we live in a constant disruption of that cycle: we eat without having felt hungry, we often ingest foods we shouldn't, without having expended energy, and we turn the reward into routine.

But somewhere, beneath layers of modern habits, our bodies still remember. They still await that ancient language. They still function, silently, as they always have: ready to alternate absence with effort, and effort with reward. Because, after all, we are not so different from those first humans who, driven by hunger and determination, ventured out into the world in search of their next meal... and found, in that process, much more than sustenance.

2.4. Stress and its impact on the body

The stress of modern life is not a fleeting noise or a passing storm; it is more like a constant drizzle that seeps in little by little, almost imperceptibly, until it soaks into every corner of daily life. It no longer stems solely from visible or immediate threats, as in ancient times, but from a complex array of worries: work responsibilities that don't end when you leave the office, family demands that require your presence and attention, financial uncertainties that weigh like a persistent shadow, and fears related to your own health or the health of those around you. It is a silent, sustained stress that is rarely completely relieved.

The human body, however, doesn't distinguish between real danger and constant worry. It responds in the same way: by activating alert mechanisms that were originally designed for brief, specific situations. Hormones like cortisol and adrenaline are released to prepare us for action, to flee or fight. But when this state persists, when there is no clear resolution or genuine rest, the body enters a kind of permanent tension that begins to erode its most delicate functions.

One of the first areas to suffer is sex life. Desire, far from being an automatic impulse, is a phenomenon deeply sensitive to physical and emotional state. For it to exist, the body needs to feel safe, relaxed, and available. But stress works in the opposite way: it prioritizes survival over pleasure, urgency over connection. The mind, preoccupied with worries, loses space for desire; the body, tense, reduces its responsiveness. Little by little, intimacy becomes secondary, displaced by tiredness, anxiety, or lack of energy.

Furthermore, the hormonal system itself is disrupted. Excess cortisol can interfere with the production of sex hormones, affecting both desire and performance. In men, this can lead to decreased testosterone; in women, to imbalances that affect the menstrual cycle and libido. But beyond the physiological aspects, there is a key emotional component: disconnection. When the mind is constantly occupied, it becomes difficult to be present, and without presence, intimacy loses its depth.

Simultaneously, sleep—that fundamental pillar of health—begins to fragment. Sleep is not simply closing your eyes; it is a complex process in which the body repairs itself, the brain organizes information, and the nervous system finds equilibrium. However, stress introduces constant interference into this process. The mind, unable to switch off, remains active even in darkness. Thoughts that were relegated to the background during the day resurface with force: to-do lists, worries, hypothetical scenarios.

Falling asleep then becomes a challenge. The body is tired, but not relaxed; the mind is exhausted, but not calm. And when sleep is finally achieved, the quality of rest is not always deep. Nighttime awakenings, light sleep, or the feeling of not having rested upon waking are manifestations of that state of alertness that refuses to switch off.

This sleep deficit, in turn, exacerbates the problem. Insufficient rest affects mood, increases irritability, reduces the ability to manage stress, and disrupts hormonal balance. This creates a vicious cycle: stress prevents good sleep, and lack of sleep increases stress.

In this context, both sex life and sleep cease to be natural and spontaneous aspects and become sensitive indicators of our internal state. They are like silent thermometers that reflect the level of balance—or imbalance—in which we live. When stress dominates, both suffer; when the body and mind find spaces of calm, they tend to restore themselves.

The paradox of modern life is that, despite having more comforts and advancements than ever before, we have created an environment in which rest and pleasure—two of the most essential functions of human beings—are constantly threatened. Recovering them doesn't necessarily mean eliminating all sources of stress, something almost impossible, but rather learning to create genuine spaces for disconnection, moments in which the body can abandon its state of alert and remember what it feels like to be calm.

Because, ultimately, sleep and intimacy are not luxuries or extras: they are deep needs, mechanisms of equilibrium that the body insistently demands. And perhaps, amidst the accelerated pace of modern life, heeding that call is one of the most important—and most forgotten—acts for preserving health.

2.5. mTOR pandemic

The reader has probably never heard this term before, since we are coining it right now, but perhaps to understand what this pandemic entails, we should explain what we mean by mTOR

<u>What is mTOR ?</u>

Deep within our cells, beyond the reach of our senses, a constant, silent, and crucial conversation for our health takes place. It's not a conversation of words, but of chemical signals that tell the body what to do at any given moment: grow, repair, store energy... or, conversely, conserve, recycle, and protect itself. Among these signals, two major metabolic “conductors” stand out. Although we don't see or feel them directly, they greatly influence how we age and how we become ill: the pathways known as AMPK and mTOR .

We can imagine these two pathways as two sides of the same coin, two complementary forces seeking balance. On one hand, AMPK acts as an energy sensor, a kind of guardian that activates when the body detects limited resources. On the other, mTOR is the great driver of growth, the one that kicks in when there is abundance and the organism can afford to build, multiply, and expand.

When we talk about AMPK, we're talking about a signal that activates in situations that, from an evolutionary perspective, were commonplace: fasting, exercise, scarcity. It's the pathway that says: "There isn't enough energy; we need to optimize, repair, eliminate what's no longer needed, and maintain what's essential." Under its influence, the body improves its efficiency, promotes fat burning, activates cellular recycling processes (like autophagy), and protects against wear and tear. It's, in a way, a pathway associated with intelligent survival.

mTOR , on the other hand, conveys the opposite message: “There is plenty, it’s time to grow.” When this pathway is activated—especially by the presence of nutrients, particularly proteins and certain amino acids, as well as insulin—the body enters building mode. Proteins are synthesized, cells divide, and tissues grow. It is a fundamental pathway for the development, regeneration, and maintenance of muscle mass. Without mTOR , we could neither grow nor recover.

So far, both seem essential, and they are. The problem lies not in their existence, but in the imbalance between them.

For most of human history, these two pathways alternated naturally. There were periods of scarcity (which activated AMPK) and periods of abundance (which activated mTOR). The body was prepared for this ebb and flow, for this rhythm. However, in modern life, this balance has been profoundly disrupted.

mTOR activation . We eat several times a day, often without having depleted our energy reserves; we continuously consume high-calorie, high-protein foods; we rarely experience prolonged fasting; and physical exercise, which would activate AMPK, is often scarce. The result is that the body receives the message to “grow, build, multiply” almost constantly.

In the short term, this may seem positive. But in the long term, the story changes.

mTOR activation means that cells are constantly stimulated to grow and divide. This sustained growth can become a problem when fine-tuning these processes is lost. Under normal conditions, the body has mechanisms to detect errors, repair damage, or eliminate defective cells. But when the growth signal is too persistent, these mechanisms can become overwhelmed.

This is where its relationship with degenerative and cancerous diseases comes in.

In the case of cancer, the link is especially clear. Cancer, in essence, is uncontrolled cell growth. And mTOR is precisely one of the pathways that promotes growth. When this pathway is overactivated , it can allow cells with mutations—which under other circumstances would have been eliminated—to continue growing and multiplying. It is not the only cause of cancer, of course, but it is a factor that can contribute to creating an environment conducive to its development.

Regarding degenerative diseases, the mechanism is more subtle, but equally important. The body periodically needs to activate internal cleaning processes, eliminating damaged or aged cellular components. This process, known as autophagy, is closely linked to the activation of AMPK and, to some extent, inhibited by mTOR . When mTOR is constantly active, this "cleaning" is reduced. It's as if a house keeps expanding and being decorated without ever taking out the trash.

Over time, this accumulation of cellular "waste" can contribute to the deterioration of tissues and organs, favoring age-related diseases such as neurodegenerative, metabolic, or cardiovascular problems.

The key, therefore, is not in eliminating mTOR or living permanently in AMPK mode, but in restoring the natural balance between both pathways. The body needs periods of growth, but also periods of repair. It needs nourishment, but also rest from food. It needs activity, but also recovery.

Practices such as physical exercise, periods of fasting, or simply spacing out meals can promote the activation of AMPK, restoring some of the body's lost balance. Similarly, a proper diet allows mTOR to perform its function in a controlled manner, without leading to constant stimulation.

Ultimately, these two paths remind us of something profoundly simple yet often forgotten: health is not a static state, but a dynamic equilibrium. It's not about always doing the same thing, but about alternating, listening, and respecting the rhythms for which we were designed.

Because, although the world has changed radically, our cells continue to function under the same old rules. And in that invisible dialogue between AMPK and mTOR , the quality and length of our lives largely depend.

<u>And why do we say it's a pandemic?</u>

Looking back at human history, major pandemics have always had something in common: they erupt visibly, generate alarm, collapse systems, and force a response. The Black Death, for example, wiped out entire populations in a matter of years; the Spanish Flu spread with devastating speed in a world that was just beginning to globalize; even the recent COVID-19 brought the planet to a standstill

and highlighted our collective vulnerability. All of them had a face, evident symptoms, daily figures, and a clear narrative of threat. However, what we might call the “ mTOR pandemic ” is of a radically different nature: it is not transmitted through the air or by contact, it does not present with fever or cough, it does not force lockdowns or immediately overwhelm hospitals... and yet, it advances much more silently, steadily, and globally.

Unlike pandemics that arrived like a violent storm, chronic activation of the mTOR pathway is more like a slow, rising tide that barely registers. It doesn't kill in days or weeks, but rather erodes over years, even decades. It has no clear beginning or end because it doesn't depend on an identifiable external agent, but on a lifestyle shared by millions: eating continuously, moving little, living in a near-permanent state of energy abundance. It's a pandemic without urgent headlines, without infection maps, without daily press conferences. And precisely for that reason, it's more difficult to perceive... and to combat.

While classic pandemics forced humanity into isolation, reduced contact, and protection from the outside world, this new pandemic, paradoxically, arises from our own everyday environment. It is not something that attacks us from the outside, but something we activate from within, every day, almost without realizing it. It doesn't generate immediate fear, but rather a false sense of normalcy. And yet, its consequences are deeply linked to many of the diseases that currently dominate mortality statistics: cancer, degenerative diseases, and metabolic disorders.

Another key difference is the perception of time. Historical pandemics were urgent: they demanded rapid responses, drastic measures, and immediate solutions. The “ mTOR pandemic ,” on the other hand, unfolds at a slow, almost imperceptible pace. Its effects are not linked to a specific cause from the previous day, but rather to an accumulation of habits over the years. This makes raising awareness extremely difficult, because humans are much better equipped to react to visible threats than to diffuse processes.

And yet, there is a profound similarity among them all: their collective nature. Just as viruses spread by exploiting interaction between people, this constant activation of mTOR extends through a shared lifestyle: the constant availability of food, widespread sedentary behavior, and a culture of immediacy. It is a cultural pandemic, almost invisible, but as global as any that have marked history.

Perhaps the greatest paradox is that, while past pandemics spurred advances in hygiene, medicine, and social organization, this new pandemic requires something different: not so much additional technology, but a conscious return to certain basic principles. It is not fought with vaccines or antibiotics, but with everyday decisions that rebalance the body's internal functioning.

Because, in the end, the “ mTOR pandemic ” isn't an external enemy we can point the finger at, but a reflection of how we live. And perhaps for that very reason, even though it's quieter and less spectacular than the great health crises in history, its impact can be just as profound... or even greater, precisely because it advances silently.

2.6. Life expectancy

When we compare the life expectancy of our ancestors with that of today, the first impression is clear and compelling: we live longer now. The figures confirm this unequivocally. Where in ancient times reaching advanced age was an almost legendary exception, in the modern world it has become commonplace. However, beneath this apparent truth lies a more complex reality, one that invites us to look beyond the numbers and ask not only how long we live, but how we live.

Our ancestors inhabited a hostile, unpredictable world where life often hung by a thread. There were no antibiotics, no vaccines, no advanced surgery. A minor infection could become a death sentence, a poorly healed wound could end a life, and diseases we now consider treatable were then fatal. Added to this was an environment in which humans didn't always occupy the top of the food chain: predators, the climate, scarcity of resources—everything conspired to shorten life. It wasn't that their bodies were more fragile, but rather that the circumstances were relentless.

Today, however, we have managed to tame many of those risks. Thanks to advances in medicine, we have learned to prevent, diagnose, and treat diseases that were once unavoidable. Hygiene, technology, scientific knowledge, and social organization have built a safety net that allows us to survive situations that would have been lethal in the past. We no longer live in constant fear of being hunted, nor do we depend solely on luck to overcome illness. In this sense, humanity has achieved one of its greatest victories: prolonging life.

But prolonging life doesn't always mean living healthier in absolute terms. This is where the comparison becomes more interesting, almost paradoxical. Because, although our ancestors lived fewer years, their bodies were deeply adapted to their environment. They were constantly moving, they ate real, unprocessed food, and they unknowingly respected the natural cycles of hunger and satiety, of activity and rest. Their physiology functioned in harmony with their lifestyle.

We, on the other hand, have created an environment that protects life, but that often challenges the body's natural balance. We live longer, yes, but we also live longer with chronic illnesses, with metabolic imbalances, with a slow decline that doesn't always manifest immediately, but accompanies many for decades. It's as if we've won the battle against premature death, but at the same time opened the door to new, more silent forms of frailty.

It is not, therefore, a matter of idealizing the past or rejecting the present. Our ancestors, if they could choose, would not hesitate to embrace the advances of modern medicine, the security of not being prey, the peace of mind of knowing that an illness is not necessarily the end. For them, all of that would be nothing short of miraculous. Likewise, we, if we look honestly, would recognize the value of their lifestyle: constant movement, connection with nature, simple and wholesome food, respect for biological rhythms.

Ultimately, each era possesses something the other longs for. They would like our protection; we, their balance.

And perhaps therein lies the key, in that kind of meeting point between two worlds. Because the combination of both—the security and knowledge of the present, along with the instinctive wisdom of the past—creates a profoundly human ideal. A way of life in which medicine protects us when necessary, but in which daily life is built on habits that honor the nature for which we were designed.

3. What is the NEXUS method? The 5 Pillars

Perhaps it's not about choosing between past and present, but about reconciling them. About living in a modern world without forgetting who we are at our core. About taking advantage of the advancements that allow us to live longer... but learning, at the same time, to live better. Because, in the end, true wealth lies not just in adding years to life, but in adding life to those years.

The NEXUS method is a comprehensive and systemic approach designed to optimize health, energy, and longevity by connecting five essential biological systems that work together to promote healthy aging. Its name is an acronym where each letter represents a fundamental pillar of human physiology: Nutrition , Exercise , Expression , Unwind , and Stress Adaptation .

The fundamental goal of this method is to "stretch life," ensuring that the additional years are of high quality by improving metabolic, cognitive, and hormonal health.

The NEXUS method is defined as the point of convergence where past habits and modern society meet to create sustainable health. It is not a list of isolated habits, but a coherent, science-based lifestyle model.

The following details what each pillar consists of and what the purpose of this method is:

3.1. Nutrition : The Real Food Revolution

Real fooding is a movement or approach to eating that promotes basing your diet on whole or minimally processed foods and minimizing the consumption of ultra-processed products . Simply put, its main idea is that your diet should focus on "real" food: fruits, vegetables, legumes, eggs, fish, meat, nuts, tubers, whole grains, and other foods that are as close as possible to their natural state. The concept was popularized in Spain by dietitian-nutritionist Carlos Ríos, who defines it as a lifestyle aimed at "eliminating ultra-processed foods from the diet " and returning to "real food."

Unlike a strict diet with fixed menus, real fooding is not usually presented as a temporary plan, but rather as a philosophy or lifestyle. In other words, it's not simply about losing weight for a few weeks, but about changing the way we understand food and making more conscious choices when shopping, cooking, and eating. Its central message is that much of the modern diet is dominated by highly processed industrial products, designed to be very appealing, easy to consume, and long-lasting, but which often have a lower nutritional quality than fresh or minimally processed foods.

Main idea of real fooding

The basis of real fooding is quite simple:

- Prioritize real or minimally processed foods
- Reduce or avoid ultra-processed foods
- Learn to read labels
- Cook more at home
- Choose food for its nutritional quality, not just for convenience or advertising.

That's why real fooding often emphasizes that it's not enough to focus solely on calories. From this perspective, consuming 300 calories from a natural and satisfying food is not the same as consuming 300 calories from a highly palatable , low-fiber, and easily overeaten processed product. In other words, this approach attempts to shift the focus from simply counting calories to the overall quality of food.

What does real fooding mean by “real food”?

Within this approach, real food refers to foods that have undergone minimal industrial processing and retain a recognizable composition. These include, for example:

- fruit
- vegetables and greens
- legumes
- tubers
- nuts and seeds
- eggs
- fish and seafood
- fresh meats
- natural yogurt with no added sugar
- simple whole grains
- extra virgin olive oil
- water
- coffee or unsweetened tea

The idea is that these foods form the basis of a healthier diet because, in general, they provide:

- more fiber
- improved nutrient density
- less added sugar
- less low-quality fat
- fewer unnecessary additives
- greater satiety capacity

This doesn't mean that all foods should be eaten raw or without any preparation. In fact, real foodism accepts that many foods are reasonably processed: for example, canned cooked legumes, frozen vegetables, canned fish, plain yogurt, or bread made with simple ingredients. In other words, processed doesn't always mean bad.

Difference between processed and ultra-processed

One of the most important points to understand real fooding is to distinguish between processed and ultra-processed food .

1. Processed foods

These are foods that have been modified in some way to make them safer, longer-lasting, or more convenient, but without losing their essential character. Some examples would be:

- frozen vegetables
- cooked legumes in a jar
- canned tuna
- natural yogurt
- olive oil
- cheese
- Traditional bread with few ingredients

In these cases, the food remains recognizable and its composition remains relatively simple.

2. Ultra-processed foods

Ultra-processed foods are industrial products made with multiple ingredients, additives, refining processes, and complex technological methods. They are typically designed to be very tasty, convenient, and have a long shelf life. Harvard describes these products as ready-to-eat foods, manufactured with several processing steps and multiple ingredients, often highly profitable and designed to encourage repeated consumption; typical examples include sugary drinks, cookies, snacks, some breakfast cereals, frozen meals, and processed meats.

Typical examples of ultra-processed foods would be:

- soft drinks
- industrial pastries
- cookies
- savory snacks
- sugary cereals
- sweetened dairy desserts
- highly elaborate commercial sauces
- highly processed sausages
- industrial fast food
- prepared meals of low nutritional quality

Real fooding relies heavily on this distinction: not everything industrial is automatically "bad", but it does consider problematic the excess of products designed more to sell and be very appealing than to nourish well.

Why did it become popular?

Real fooding became popular because it offers a very easy-to-understand message:

Eat real food and reduce ultra-processed foods .

That simplicity allowed many people to quickly connect with the concept. In contrast to complicated diets, calorie counting, or difficult rules, real fooding proposes something seemingly more intuitive:

- back to basics
- cook more
- read labels
- be wary of food marketing
- prioritize simple foods

Furthermore, its success is also due to the fact that it responds to a fairly widespread social feeling: the idea that nowadays there are too many food products wrapped in healthy advertising ("light", "fitness", "natural", "no added sugar", etc.) that in reality are still not recommended if consumed frequently.

The movement gained a lot of visibility with the book Eat Real Food, which presents the foundations of real fooding as a lifestyle that seeks to transform eating habits and reduce the presence of ultra-processed foods .

<u>Basic principles of real fooding</u>

Although it may vary depending on who explains it, the most common principles are:

1. Base your diet on real food

Most of the diet should come from fresh or minimally processed foods.

2. Reduce consumption of ultra-processed foods

It is not always presented as an absolute prohibition, but rather as a very clear and conscious reduction.

3. Learn to shop better

Real fooding places great importance on:

- read ingredients
- Don't be fooled by the packaging
- understand the labeling
- distinguish advertising of real quality

4. Cook more at home

Cooking allows you to:

- control ingredients
- reduce hidden sugars and fats
- adjust portions
- improve your relationship with food

5. Prioritize satiety and nutrition

Value is placed on foods that truly nourish, satisfy, and provide nutrients, instead of products designed to be very appealing but not very satisfying.

<u>Benefits of real fooding</u>

1. Improvement of the overall quality of the diet

That's probably its biggest advantage. If a person replaces:

- pastries instead of fruit and natural yogurt
- soft drinks for water
- snacks made with nuts
- dishes prepared with homemade food

It is very likely that their diet will improve significantly.

2. Reduction in the consumption of ultra-processed foods

This could have a positive impact, because greater exposure to ultra-processed foods has been associated with worse health outcomes. A large review published in *The BMJ* found that greater exposure to ultra-processed foods was linked to a higher risk of several health problems, particularly cardiometabolic disorders , type 2 diabetes, and cardiovascular mortality, although the quality of the evidence varies depending on the outcome.

3. Greater satiety and better appetite control

Real food usually provides more:

- fiber
- protein
- water
- volume

This can help you feel more satiated and eat less impulsively.

4. Better nutrition education

Real fooding , when properly understood, can teach people to:

- interpret labels
- recognize marketing strategies
- value simple ingredients
- to differentiate between “seems healthy” and “is healthy”

5. Promoting sustainable habits

It can also encourage habits such as:

- cook more
- plan purchases
- eat more slowly
- depend less on ready-to-eat products

In summary, this pillar seeks to escape the "matrix" of ultra-processed foods in order to eliminate chronic systemic inflammation, which is the root of most modern diseases.

- Restoring biological control: By consuming foods that maintain their original matrix, the "hijacking" of brain dopamine caused by products designed to be addictive is avoided (the *bliss)* . *point*), allowing natural satiety signals (leptin) to function again.
- Digestive and mental health: Consuming prebiotic fiber nourishes the microbiota , promoting the production of up to 90% of serotonin in the gut, which directly impacts mood and the immune system.
- Metabolic stability: By eliminating liquid sugars and refined flours, constant insulin spikes are avoided, reducing the risk of diabetes and promoting stable energy throughout the day.

3.2. Exercise : Muscle as life insurance

Heavy weight training is a form of strength or endurance training that uses weights that are heavy relative to a person's maximum lifting capacity. Its main goal is to stimulate the muscular and nervous systems to produce adaptations such as increased strength, power, muscle mass, and improved bodily functions. Simply put, it involves performing exercises with a load heavy enough to make the effort intense, while the number of repetitions is relatively low or moderate.

In the field of training, when we talk about "heavy loads" we usually refer to weights that represent a high percentage of the 1RM (one repetition One-repetition maximum (1RM), that is, the maximum weight a person can lift once with good technique. Generally, loads approximately above 70–80% of the 1RM are considered heavy, although this can vary depending on the exercise, the individual's fitness level, and the training goal. The American College of Sports Medicine (ACSM) has stated for years that moderate-to-high to heavy loads are typically used to develop strength in healthy adults, and that progressions should be tailored to each person's fitness level.

Unlike lighter training, which uses small weights and many repetitions, heavy weight training is characterized by:

- greater mechanical tension on the muscle
- greater demand on the nervous system
- fewer repetitions per set
- greater need for technique and control
- greater importance of rest and recovery

Therefore, it is not simply about "lifting a lot of weight", but about doing it in a structured, progressive and safe way, respecting technique and individual level.

<u>What exactly is heavy weight training?</u>

Heavy weight training is part of strength training. It can be performed with:

- free weights (dumbbells, barbells, kettlebells)
- gym machines
- pulleys
- sleds or resistance implements
- even body weight itself, when the difficulty is high enough

What defines "heavy" is not the object itself, but the relative effort it requires from the person. For example, a 20 kg dumbbell might be very light for one person and very heavy for another. Therefore, the concept always depends on individual capacity.

In practice, this type of training usually includes exercises such as:

- squats
- dead weight
- bench press
- military press
- rowing
- weighted dominated
- hip thrust
- heavy strides
- multi-joint exercises on machines

These are exercises that allow you to move a lot of weight and involve large muscle groups, so they generate a very powerful overall response.

<u>How does it work in the body?</u>

When a person trains with heavy loads, the body receives a stimulus of high mechanical and neuromuscular demand. To respond to this stimulus, a series of adaptations occur:

1. Increase in mechanical stress

The heavy load forces the muscle to generate a lot of force to move the weight. This tension is one of the most important signals for the body to adapt.

2. Recruitment of more muscle fibers

With heavy loads, the body needs to activate a greater number of motor units, especially the fast-twitch muscle fibers, which have the greatest potential to produce force and power.

3. Adaptations of the nervous system

Heavy training doesn't just work the muscles; it also improves the nervous system's ability to:

- activate more fibers at once
- coordinate them better
- generate force faster
- improve technique under pressure

4. Adaptive response

Over time, the body becomes more efficient and stronger, because it understands that it needs to be prepared to withstand that repeated demand.

Main benefits of heavy weight training

1. Increase in maximum strength

clearest and most characteristic benefit . Heavy weight training is especially effective for improving maximum strength, that is, the ability to produce the greatest possible amount of force in a single effort.

This happens because:

- improves neuromuscular recruitment
- increases coordination between muscles
- improves technical efficiency
- increases the ability to apply real force against high resistance

The scientific evidence is quite clear on this point: when comparing high loads with low or very low loads, high loads generally produce greater improvements in maximal strength, especially in tests like the 1RM. Several systematic reviews and meta-analyses have found that high loads improve maximal dynamic strength more than light loads, even when the total volume is similar.

2. Increase in muscle mass (hypertrophy)

Although many people think that heavy weights are only for strength and not for "building muscle," that's not correct. Training with heavy weights can also produce muscle hypertrophy, that is, an increase in muscle size.

High loads generate:

- high mechanical stress
- high muscle activation
- strong stimulus for adaptation

Interestingly, current research shows that hypertrophy can be achieved with different load ranges, provided the training is done correctly and with sufficient effort. In other words:

- Heavy weights are great for gaining muscle
- But they are not the only way to achieve it
- Light loads can also work if they get very close to failure

Even so, heavy loads have the advantage of building muscle while more significantly improving maximum strength. Scientific reviews indicate that muscle growth can be similar with high or low loads when the effort is sufficient, but maximum strength typically improves more with heavy loads.

3. Improved bone density

One of the most important, and sometimes lesser-known, benefits is that heavy weight training can help improve or maintain bone mineral density.

When bones are subjected to repeated mechanical loads:

- The bone tissue receives an adaptation signal
- The body understands that it needs stronger bones.
- bone remodeling is stimulated

This is especially important for:

- prevent osteopenia
- reduce risk of osteoporosis
- improve skeletal strength
- reduce the risk of fractures in the long term

Mayo Clinic highlights that well-executed weight training helps improve bone density, and this is one of the reasons why it is recommended for adults and older people, always with appropriate progression.

4. Improvement of body composition

Heavy weight training can greatly improve body composition, that is, the relationship between:

- muscle mass
- fat mass
- total body mass

This happens because:

- It promotes the maintenance or increase of muscle.
- It can help reduce fat if the diet is accompanied by proper nutrition.
- improves the "functional physical appearance" of the body
- increases the energy expenditure associated with muscle tissue

It doesn't mean it magically "burns fat on its own," but it does help the body to:

- more lean mass
- improved muscle tone
- improved metabolic efficiency
- greater ease in maintaining long-term results

5. Improved metabolism and insulin sensitivity

Muscle is a metabolically very important tissue. The more functional muscle mass a person has, the better their body's ability to:

- manage glucose
- store glycogen
- respond to insulin
- use energy efficiently

Therefore, strength training, including working with heavy loads, can help to:

- improve insulin sensitivity
- promote glycemic control
- reduce metabolic risk
- support the prevention of problems such as insulin resistance or type 2 diabetes (within a healthy lifestyle)

6. Joint protection and improved stability

Although many people believe that "heavy weight damages the joints," in reality, when done correctly, heavy weight training can strengthen structures that protect the joints.

Help to improve:

- strength of stabilizing muscles
- motor control
- support around knees, hips, shoulders and spine
- coordination and posture

That can be translated into:

- lower risk of injury
- improved tolerance to daily stress
- Improved ability to bend, lift objects, push, or carry

However, the benefits only appear when there is good technique, proper progression, and correct exercise selection. The Mayo Clinic emphasizes that weight training offers clear benefits, but poor technique increases the risk of sprains, strains, fractures, and other injuries.

Functional benefits in daily life

Heavy weight training isn't just for athletes or bodybuilders. It also improves performance in everyday activities such as:

- climb stairs
- carry shopping bags
- lift weight from the floor
- move more safely
- maintain balance
- prevent falls
- Maintaining independence with age

In adults and older people, this is especially important because it helps to combat:

- age-related loss of muscle mass
- fragility
- sarcopenia
- loss of functional strength
- progressive physical dependence

Therefore, heavy or relatively challenging strength training, adapted to each person, is a very useful tool for healthy aging.

Benefits to the nervous system and performance

Heavy weight training greatly improves the nervous system. This is key because strength depends not only on muscle size, but also on how the brain and nerves "direct" movement.

This type of work improves:

- neuromuscular activation
- intramuscular coordination (within the same muscle)
- intermuscular coordination (between muscles)
- recruitment speed
- efficiency of the technical pattern

Therefore, it usually increases:

- the ability to produce force quickly
- the power
- stability
- body control

This benefits not only those looking to lift heavier weights, but also:

- field athletes
- people who practice explosive sports
- people who want to improve overall performance

NEXUS understands muscle not as an aesthetic system, but as the largest endocrine organ in the body.

- Slowing down aging (Sarcopenia): Strength training is the only tool capable of stopping the natural loss of muscle mass that accelerates from the age of 30.
- Bone and Metabolic Health: Through axial loading (such as squats), Wolff's law is activated, strengthening bone mineral density against osteoporosis and osteopenia. Furthermore, muscle acts as the primary glucose storage site, drastically improving insulin sensitivity.
- Systemic signaling: When it contracts, the muscle releases myokines , molecules that travel through the blood communicating with the brain, liver, and pancreas to improve overall health.

3.3. Vitality (eXpression): Sexual health as a biomarker

Having an active sex life means maintaining frequent or regular sexual activity, voluntarily, healthily, and in accordance with the desires, needs, and circumstances of each person or couple. There is no universally "correct" number of sexual encounters, because the frequency considered active can vary greatly depending on age, health status, emotional state, whether or not a person is in a relationship, sexual orientation, level of sexual desire, and lifestyle. Therefore, rather than measuring it solely by "how many times," an active sex life is generally understood as one in which a person experiences their sexuality in a satisfying, safe, and physically and emotionally fulfilling way.

From a health perspective, sexuality is not limited solely to the sexual act itself. It also includes:

- sexual desire
- excitement
- privacy
- keen
- pleasure
- physical contact
- orgasm
- emotional connection
- satisfaction with intimate life

In fact, sexual health is considered an important part of overall well-being. The NHS (the UK's National Health Service) states that sexual health is part of global health and encompasses physical, emotional, and relational well-being related to sexuality.

Therefore, when we talk about an "active sex life," we are not only referring to having penetrative sex, but to a present, healthy, and satisfying intimate life, always based on:

- consent
- I respect
- security
- mutual desire (when there is a couple)
- absence of coercion or violence

What exactly is meant by an active sex life?

In general terms, an active sex life can include different forms of sexual expression, for example:

- penetrative sex
- oral sex
- intimate caresses
- kisses and erotic contact
- individual or shared masturbation
- consensual sex games
- physical and emotional intimacy

This is important because sexuality is often reduced solely to intercourse, when in reality a healthy and active sex life can include many forms of pleasure and connection.

Furthermore, an active sex life does not necessarily mean having a partner. A person can have an active sex life through:

- relationships with a stable partner
- consensual casual relationships
- self-exploration or masturbation
- Intimate life not focused solely on intercourse, but also on emotional and physical connection

From a broader perspective, what matters is not just the frequency, but that sexuality is experienced in a way that contributes to well-being and does not cause physical or emotional harm.

What happens in the body during sexual activity?

Sexual activity produces a combination of physical, hormonal, neurological, and emotional responses.

During arousal and orgasm, various body systems are usually activated:

1. Cardiovascular system

- Heart rate increases
- Blood flow increases
- It improves blood flow to certain areas of the body
- Moderate physical exertion occurs in many situations

2. Hormonal and neurological system

During sexual activity, and especially during orgasm, the body can release substances related to pleasure and well-being, such as:

- dopamine (associated with pleasure and motivation)
- oxytocin (related to bonding and relaxation)
- endorphins (the body's natural painkillers)

Cleveland Clinic explains that, after orgasm, oxytocin and other substances associated with relaxation, improved mood, and, in some people, better sleep are released.

3. Muscular system

- Muscles of the trunk, pelvis, and limbs are activated.
- There may be involuntary muscle contractions
- Pelvic floor muscles are worked in some contexts

4. Emotional and psychological system

- It can increase the feeling of connection
- It can temporarily reduce stress
- It can promote feelings of intimacy, satisfaction, or relaxation

Therefore, sexual activity is not only a reproductive or intimate act, but also an experience that involves the entire organism.

Main benefits of having an active sex life

1. Improvement of emotional well-being and mood

One of the best-known benefits of an active sex life is its potential positive effect on mood. Many people experience the following after sexual activity:

- feeling of relaxation
- greater well-being
- reduction of emotional tension
- feeling of pleasure
- increased emotional closeness

This is related to the release of neurotransmitters and hormones involved in pleasure and attachment, such as dopamine and oxytocin. The Cleveland Clinic notes that these hormonal responses can help improve mood and reduce stress in some people.

Furthermore, observational studies have shown that sexually active people, especially in adulthood and older age, may exhibit a greater sense of well-being or enjoyment of life, although this does not automatically mean that one causes the other in every case. One study in older adults found that those who reported sexual activity in the past year had higher levels of enjoyment of life than those who did not.

2. Stress reduction and improved relaxation

Sexual activity can help reduce stress for several reasons:

- It reduces accumulated physical tension
- promotes subsequent relaxation
- It can lower mental activation in some people.
- It can improve the feeling of closeness and emotional security.

After orgasm, many people experience a distinct physical and emotional release. The Cleveland Clinic notes that sex and intimacy can help bring cortisol (the stress hormone) back to normal levels in some situations.

This does not mean that sex is a "therapy" for psychological problems, but it can be part of a healthy life and help some people better manage everyday stress.

3. Improved sleep

Another common benefit is better sleep after sexual activity, especially when there is orgasm.

This relates to:

- muscle relaxation
- oxytocin release
- feeling of satisfaction
- decreased state of alertness

Many people experience drowsiness or easier rest after sexual activity. The Cleveland Clinic notes that the release of oxytocin after orgasm can promote sleep and rest.

Better sleep, in turn, can have a positive impact on:

- mood
- physical recovery
- daily energy
- hormonal health

4. Cardiovascular benefits and moderate physical movement

Sexual activity can represent, for many people, a form of moderate physical activity, although it does not replace structured exercise.

During sex there may be:

- increased heart rate
- energy expenditure
- increased blood circulation
- mobilization of several muscle groups

Cleveland Clinic describes sex as a form of moderate physical activity in some cases, although it notes that it does not replace regular exercise recommended for health.

This can contribute, in a modest way, to:

- maintain mobility
- improve circulation
- stimulate the cardiovascular system
- increase the feeling of vitality

However, it should not be considered a substitute for physical training, but rather a complement within an active lifestyle.

5. Strengthening the couple bond and emotional intimacy

When there is a partner, an active and satisfying sex life can help to:

- increase emotional connection
- strengthen intimacy
- improve body language
- increase the feeling of closeness
- strengthen the emotional bond

Not all relationships work the same way, nor is sexuality the only important factor, but a healthy intimate life can be a relevant part of a couple's well-being.

Sexuality, in that sense, can act as a form of:

- expression of affection
- complicity
- mutual validation
- maintaining desire and connection

Specific physical benefits in the body

1. Activation and maintenance of the pelvic floor

In many people, regular sexual activity can involve pelvic muscle work and increased body awareness of the genital area.

This may be related to:

- improved body awareness
- better sexual response
- greater comfort in intimacy
- In some cases, improved pelvic floor tone

Cleveland Clinic mentions that, in some people, regular sexual activity can help strengthen the pelvic floor and reduce discomfort during intercourse, especially when there is good adaptation and absence of pain.

2. Possible pain relief in some situations

In some cases, some people experience a temporary decrease in pain after orgasm, due to the release of endorphins.

It has been described that it can help, in some people, to provide temporary relief:

- muscle tension
- headache
- accumulated physical stress
- mild discomfort

Cleveland Clinic indicates that endorphins released during sex can act as natural painkillers and that some people report relief from headaches or migraines.

This is not universal nor does it replace medical treatment, but it is an effect that is observed in some people.

3. Possible long-term effects on sexual function

Maintaining an active sex life can be beneficial for some people:

- better knowledge of one's own body
- greater ease in identifying desire and arousal
- better sexual response
- more confidence in intimacy
- greater sexual satisfaction

Masturbation and self-exploration can also be part of a healthy sexuality. The Cleveland Clinic notes that masturbation can help reduce stress, improve mood, facilitate sleep, and, in some people, improve their sex life by providing a better understanding of their own body and its response to pleasure.

Psychological and self-esteem benefits

An active and satisfying sex life can also have a positive influence on:

- self-esteem
- body perception
- personal safety
- feeling of attractiveness
- trust in intimacy
- perception of connection with oneself or with one's partner

This does not mean that personal worth depends on sexual activity, but it does mean that a healthy sexuality can contribute to a better perception of one's own well-being.

In older adults, Mayo Clinic notes that maintaining sexual interest, satisfaction with the frequency, and enjoyment of sexual life is associated with good health and well-being in old age.

In general, this pillar uses sexual activity as a powerful neurobiological and hormonal regulator.

- Masterful stress management: The release of oxytocin during intimacy directly counteracts cortisol, which promotes muscle retention and reduces abdominal fat accumulation.
- Hormonal support and anti-aging benefits: Sexual activity can temporarily increase DHEA, a hormone precursor that improves immune function and bone density. In men, ejaculatory frequency is associated with a lower risk of prostate cancer.
- Cardiovascular Health: Improves endothelial function (the lining of blood vessels), facilitating better blood flow and reducing the risk of cardiovascular disease.

3.4. Recovery (Unwind): The biological mandate of sleep

Sleep hygiene is the set of habits, routines, and environmental conditions that promote good quality sleep that is sufficient in duration and restorative for the body and mind. In other words, it involves a series of behaviors that help you sleep better, fall asleep more easily, stay asleep throughout the night, and wake up feeling adequately rested.

It doesn't refer to "hygiene" in the sense of physical cleanliness, but rather to taking care of how we relate to rest. It includes aspects such as:

- maintain regular bedtimes and wake-up times
- create a suitable environment for sleeping
- Avoid stimulants before bed
- reduce screen time at night
- control naps
- take care of your diet and physical activity
- establish relaxation routines before bedtime

The central idea is that sleep depends not only on "feeling sleepy," but also on how we organize our day, our environment, and our habits. Both public health organizations and medical centers agree that improving sleep habits can significantly contribute to better sleep. The Centers for Disease Control and Prevention recommends, for example, going to bed and waking up at the same time, keeping the room cool, quiet and dark, turning off electronic devices at least 30 minutes before sleeping and avoiding heavy meals, alcohol and caffeine close to nighttime.

Therefore, sleep hygiene is not a medication or an isolated technique, but a lifestyle strategy that seeks to protect rest as an essential biological function.

<u>What exactly is sleep hygiene?</u>

Sleep hygiene can be defined as a set of healthy practices aimed at optimizing sleep quality. Its goal is not just to sleep more hours, but to sleep better.

Sleeping well involves several things:

- to be able to fall asleep in a reasonable amount of time
- sleep relatively continuously
- achieve restful sleep
- waking up feeling rested
- function well during the day

The Centers for Disease Control and Prevention explains that quality sleep means that sleep is continuous and restorative, not simply that the person has spent many hours in bed.

Therefore, good sleep hygiene aims to improve both the quantity and quality of rest.

<u>Basic principles of good sleep hygiene</u>

Although it may vary slightly from person to person, the most important pillars are usually the following:

1. Maintain a regular schedule

One of the most important habits is:

- go to bed at a similar time each night
- get up at a similar time every morning
- maintain a certain consistency even on weekends

This helps reinforce the circadian rhythm, which is the internal biological clock that regulates sleep-wake cycles. The Mayo Clinic specifically recommends maintaining a consistent schedule, as this strengthens the sleep-wake cycle.

2. Create a suitable sleeping environment

The bedroom should be a place that promotes rest. In general, it is recommended that it be:

- dark
- silent
- cool or lukewarm
- comfortable
- tidy
- associated with rest, not work or entertainment

The Mayo Clinic and the Centers for Disease Control and Prevention highlights that a cool, dark, and quiet room facilitates falling and staying asleep.

3. Reduce screen time before bed

One of the most repeated pieces of advice today is to avoid prolonged use of:

- mobile
- tablet
- computer
- television

just before going to bed.

This is because the light emitted by screens can make it difficult for the brain to enter rest mode, in addition to keeping the mind active through the content. The Centers for Disease Control and Prevention recommends turning off electronic devices at least 30 minutes before bed, and the Mayo Clinic also advises avoiding screens right before going to sleep.

4. Avoid stimulants and heavy meals

Sleep hygiene also includes being mindful of what you consume in the afternoon or evening. The following is generally advised:

- Avoid caffeine late
- Avoid nicotine before bed
- moderate or avoid alcohol at night
- Avoid very heavy or large dinners
- Do not go to bed intensely hungry or with a heavy digestive system.

The Mayo Clinic notes that caffeine and nicotine can interfere with sleep for hours, and that alcohol, although it may cause drowsiness at first, usually worsens sleep quality later in the night.

5. Control naps

Taking long or very late naps can make it harder to sleep well at night. Therefore, when trying to improve nighttime sleep, the following is often recommended:

- short naps
- Avoid very late naps
- Do not use them as a constant substitute if they disrupt your main sleep.

The Mayo Clinic recommends limiting long naps and avoiding sleeping late during the day when this impairs nighttime rest.

6. Include exercise and stress management

Regular physical activity usually promotes better sleep, and practices such as the following also help:

- deep breathing
- meditation
- quiet reading
- gentle stretches
- write in a diary
- slow down your mental pace before bed

The Mayo Clinic indicates that regular exercise can improve sleep, although it is advisable to avoid intense activity too close to bedtime if it overstimulates the person.

Why is sleep hygiene so important?

Sleep hygiene is important because sleep is not a "luxury" or simply passive rest. While we sleep, the body performs functions essential for health.

The National Heart, Lung, and Blood Institute explains that during sleep the body works to support brain function, physical health, tissue repair, hormonal balance, immune response, and daytime functioning.

In other words, sleeping well affects almost all of the body's systems.

Main benefits of good sleep hygiene on the body

1. Improved physical rest and body recovery

One of the most direct benefits is that it promotes more restful sleep, which allows the body to:

- better recover from daily wear and tear
- repair tissues
- regulate physiological processes better
- reduce accumulated fatigue
- wake up with more energy

The National Heart, Lung, and Blood Institute notes that sleep helps repair cells and tissues and supports overall physical health.

Therefore, good sleep hygiene can be translated into:

- less daytime fatigue
- improved muscle recovery
- less feeling of exhaustion
- improved everyday physical performance

2. Improved brain function, memory, and concentration

Sleeping well has a very important effect on the brain. Good sleep hygiene promotes deeper, more continuous sleep, and this helps to:

- consolidate memory
- learn better
- improve care
- make better decisions
- react faster
- perform better in studies or work

The National Heart, Lung, and Blood Institute explains that getting enough sleep improves learning, memory, problem-solving, attention, and decision-making.

The Centers for Disease Control and Prevention also indicate that getting enough sleep improves attention and memory, allowing you to perform daily activities better.

3. Mood regulation and stress reduction

Good sleep hygiene also greatly influences emotional health. When you sleep well, you usually experience:

- more emotional stability
- less irritability
- improved stress tolerance
- less feeling of mental overload
- improved overall mood

The Centers for Disease Control and Prevention indicates that getting enough sleep helps reduce stress and improve mood.

In addition, the National Heart, Lung, and Blood Institute warns that lack of sleep can affect emotional control, behavior, and the ability to cope with change, and has even been linked to a higher risk of depression and other psychological problems.

4. Improved immune system

Sleeping well also helps the immune system. Good sleep hygiene promotes consistent rest, and that can contribute to:

- get sick less often
- better respond to infections
- better regulate inflammation
- improve the body's natural defenses

The Centers for Disease Control and Prevention states that getting enough sleep can help you get sick less often.

In addition, the National Heart, Lung, and Blood Institute published results linking a consistent night's sleep with better immune function and less alteration of inflammatory markers when sleep is not chronically cut short.

5. Improved metabolic health and appetite control

Sleeping well not only affects rest, but also metabolism. Good sleep hygiene can help regulate:

- hunger
- satiety
- blood glucose
- the insulin response
- energy during the day

The National Heart, Lung, and Blood Institute explains that sleep influences hormones related to appetite, such as ghrelin and leptin, and that lack of sleep can disrupt glucose levels and increase the risk of metabolic problems.

That's why the Centers for Disease Control and Prevention indicates that getting enough sleep helps maintain a healthy weight and improves metabolic health.

6. Cardiovascular protection

Sleep quality also influences the cardiovascular system. Maintaining good sleep hygiene and getting adequate sleep can contribute to:

- better blood pressure regulation
- less physiological overload
- improved recovery of the cardiovascular system
- lower long-term risk of cardiometabolic disorders

The Centers for Disease Control and Prevention indicates that getting enough sleep helps improve heart health and is associated with a lower risk of hypertension, heart disease, and stroke.

The National Heart, Lung, and Blood Institute also notes that sleep helps "repair" the heart and blood vessels, and that lack of sleep increases the risk of hypertension, obesity, heart disease, and stroke.

Functional benefits in daily life

Good sleep hygiene not only improves internal parameters but also daily life. Sleeping well usually translates into:

- more energy upon waking
- improved academic or work performance
- fewer mistakes due to carelessness
- better reaction time
- more patience
- better mood
- greater capacity to train or move
- lower risk of accidents

The Centers for Disease Control and Prevention warns that getting enough sleep also reduces the risk of traffic accidents and sleep-related injuries.

In short, sleep is treated as a critical performance tool, not a luxury.

- Brain cleansing (Glymphatic System): During deep sleep, the brain activates a "washing cycle" of cerebrospinal fluid that removes metabolic waste, such as beta-amyloid protein, linked to Alzheimer's.
- Regulation of body composition: Sleep regulates appetite hormones (ghrelin and leptin); a lack of sleep makes us crave sugary and processed foods. Furthermore, most of the growth hormone (HGH), vital for cell repair, is released during deep sleep.
- Cognitive consolidation: This is the moment when the brain processes short-term memory and regulates emotions, reducing anxiety and brain fog.

3.5. Stress adaptation : Intermittent fasting

Intermittent fasting is a way of organizing your diet based on alternating periods of eating with periods of no calorie intake or significantly reduced intake. Unlike a traditional diet, it focuses less on what foods are consumed and more on when they are consumed. Therefore, rather than a strict diet, it is usually considered a strategy for distributing meal times.

The main idea behind this theory is that the body goes through different metabolic states depending on the time elapsed since the last meal. After eating, the body uses the energy from food, especially glucose, and there is increased activity of insulin, a hormone related to energy use and storage. However, when several hours pass without food intake, insulin levels tend to drop, and the body begins to draw first on its glycogen reserves and, later, on stored fat. During longer periods of fasting, the production of ketone bodies, another source of energy, can also increase.

Therefore, intermittent fasting aims to create repeated periods of digestive and metabolic rest, encouraging the body to alternate between using the immediate energy from food and utilizing its reserves. This shift is one of the reasons why it is associated with weight management and certain metabolic improvements in some people.

<u>Main types of intermittent fasting</u>

There are different ways to practice it, although the three best known are:

1. Time-restricted feeding

This is the most popular method. It consists of eating only within a specific daily time frame and fasting the rest of the day. For example:

- 12/12: 12 hours of fasting and 12 hours of eating
- 14/10: 14 hours of fasting and 10 hours of food
- 16/8: 16 hours of fasting and 8 hours of eating
- 18/6: 18 hours of fasting and 6 hours of eating

In this model, a person can choose to eat, for example, between 10:00 and 18:00, and fast from 18:00 until the following morning. It is the simplest method and usually the easiest to manage, since part of the fasting period coincides with sleep hours.

2. Alternate-day fasting

In this case, days of normal eating are alternated with days of fasting or semi-fasting. This can be done in two ways:

- Alternating complete fasting: one day you eat normally and the next day you don't eat.
- Modified alternate-day fasting: one day you eat normally and the next day you consume a very reduced amount of calories.

It is a more demanding modality and, for many people, more difficult to maintain in the long term.

3. 5:2 Method

It consists of eating normally for five days a week and fasting or severely restricting calories for two days (preferably not consecutive). It is an intermediate option between daily fasting and alternate-day fasting.

Why is it used?

Intermittent fasting has become popular because, for many people, it's a simpler way to control their food intake. Instead of constantly counting calories or weighing food, you follow a time-based rule: "I eat at this time; I don't eat at this time." This simplicity can make it easier for some people to manage than a traditional diet.

Furthermore, by reducing the eating window, many of the following are often eliminated:

- unnecessary snacking
- late-night snacks
- impulsive eating
- excess calories spread throughout the day

In other words, one of the reasons it might work is that many people end up eating less without realizing it, even if they don't count calories.

There are also those who value it because they feel it benefits them:

- more structure
- fewer food decisions
- greater sense of control
- less dependence on “snacking between meals”

However, it's important to understand that intermittent fasting isn't magic. Its effectiveness depends not only on the time spent without eating, but also on the overall set of habits.

Observed benefits

Among the most frequently mentioned benefits of intermittent fasting are the following:

1. Weight control and fat loss

This is the most common reason why people try it. It can help with weight loss because:

- reduces the frequency of intake
- snacking decreases
- facilitates a calorie deficit
- It improves control over food in some people

Even so, this doesn't automatically mean it's superior to other diets. In many cases, weight loss is explained by consuming fewer calories overall.

2. Metabolic improvements

These effects are not the same for everyone, but in general, intermittent fasting can promote:

- better glucose control
- improved insulin sensitivity
- triglyceride reduction
- improvement of some cholesterol parameters
- slight decrease in blood pressure
- reduction of abdominal circumference

Through the stress it is subjected to, the body is forced to change its metabolic "software".

- Autophagy (Cellular Recycling): This is the star benefit of this pillar; the body begins to "eat" its own damaged parts or misfolded proteins, performing a deep internal cleaning that prevents degenerative diseases.
- Metabolic Flexibility: The body learns to use fats (ketone bodies) as fuel instead of relying solely on glucose, providing a more constant and powerful energy flow to the brain.
- Cardiovascular Health and Neuroplasticity : Reduces triglycerides, improves blood pressure, and increases BDNF, a protein that acts as a "fertilizer" for neurons, promoting new synaptic connections.

3. Less snacking, better meal organization, and mental clarity

This habit is notable for its ability to curb impulsive eating driven by boredom or anxiety, resulting in an immediate positive impact on health. Those who practice it report significant physical and cognitive benefits, such as easier digestion, greater mental clarity, reduced lethargy after eating, and better adherence to their daily routines.

4. What is the NEXUS method used for?

The ultimate goal of the method is to "stretch life" (stretch) lifetime) and ensure that the additional years are of high quality. It is specifically used for:

- Improve body composition and metabolic health: Targets visceral fat, reverses insulin resistance, and protects lean tissue.
- Promote mental clarity: By stabilizing glucose and using ketone bodies as " super fuel " for the brain, it eliminates brain fog and improves neuroplasticity .
- Regulate the endocrine system and reduce stress: Balance hormones such as cortisol, leptin, and testosterone through rest and sexual vitality.
- Preventing degenerative diseases: Activating autophagy and reducing chronic inflammation are key to preventing age-related pathologies such as Alzheimer's or cardiovascular problems.
- Creating a sustainable life framework: Unlike extreme diets, NEXUS offers an adaptable "operating system" where the profile defines the approach, but the person adjusts the parameters according to their age, gender and life context.

The NEXUS method doesn't seek temporary changes, but rather a profound biological adaptation. By integrating these pillars, the individual achieves a synergy where fasting enhances nutrition, strength improves metabolism, sleep repairs the damage from training, and sexual vitality balances the stress generated by modern life, resulting in sustainable health and active longevity.

The objective of this method is not to explain the benefits of each of the pillars, since there is already a lot of literature on them, but to show the most correct way to combine them: nourishing, stimulating, expressing, recovering and adapting the biology of the individual to reach their best possible version at any stage of life.

5. NEXUS Performance Profile Definition

Based on the answers you, dear reader, have given in the completed questionnaire, you have been classified into this specific profile.

The NEXUS Performance profile (also known as the "High Performance" profile) represents an individual who has transcended basic healthcare, making it a central pillar of their lifestyle and personal performance. This archetype typically falls within the 25-50 age range and is characterized by biological maturity and a very high level of responsiveness to stimuli.

This profile is then defined in as much detail as possible, based on its behavioral, psychological, and operational characteristics:

1. Life context and structure management

Unlike other more reactive or intuitive profiles, the Performance individual operates within a well-structured environment. Their daily life isn't marked by improvisation or the feeling of constantly "putting out fires"; on the contrary, they plan their activities and training as an integral part of their schedule. They have an organized life that allows them to dedicate time and resources to achieving specific goals without this becoming an unmanageable mental burden.

2. Physiological state and energy profile

Biologically, this profile presents the following characteristics:

- Responsiveness: Your body responds efficiently to external stimuli, allowing you to progress steadily.
- Energy level: It presents a high and fairly stable energy level throughout the day.
- Recovery: He has consolidated habits that allow him to maintain a solid rest base, generally managing to sleep between 7 and 8 hours consistently.

3. Psychological profile and relationship with discomfort

The most distinctive psychological trait of this profile is its high tolerance and acceptance of discomfort. It does not seek immediate gratification nor avoid effort; it understands that discomfort is a necessary component for growth and improved performance. It possesses a strong mental focus and an established discipline that allows it to maintain consistency even during periods of high work or personal demands.

4. Motivation and strategic objectives

Their goals are ambitious and go far beyond simply "feeling better" or preventing illness. Their primary motivation is optimizing physical and mental performance, as well as pursuing active longevity. They have clear objectives and seek tangible metrics that demonstrate their progress, both in terms of physical fitness and vital energy.

5. Decision-making and self-management

Regarding its relationship with knowledge and discipline, the Performance profile is defined by:

- Empiricism and experience: When making decisions about your health, you don't rely solely on intuition or external theories; you use your own previous experience and the results obtained as a compass to adjust your habits.
- Commitment to the process: Deeply understands the biological mechanisms of the body and accepts the necessary discipline as part of their identity.
- Self-control: Possesses a much higher than average level of stress control, allowing him to apply more demanding approaches without compromising his hormonal or emotional balance.

6. Relationship with physical activity and vitality

This individual does not view exercise as sporadic or a tedious obligation. Their relationship with training is high-frequency, typically dedicating between four and six days a week to planned physical activity. If they stop training for a week, they notice a clear change in their well-being and performance, which reinforces their commitment to consistency. Furthermore, they consider vitality a key part of their daily balance and energy.

In summary, NEXUS Performance defines a highly disciplined, proactive, and results-oriented person who sees the method as a tool to elevate their biological potential to the highest possible level.

6. Objectives for the NEXUS Performance profile

For the NEXUS Performance profile, the method's objectives transcend basic health to focus on maximizing biological potential, physical and mental performance, and building elite active longevity . This individual has a high tolerance for discomfort and seeks tangible metrics to support their progress.

The following details the strategic objectives that the system seeks to achieve for this profile:

1. Achieve superior metabolic efficiency and mental acuity (Pilar S)

- The central objective is to master the 20-4 protocol (Warrior Style) to transform the body's response to hormetic stress .
- Maximizing autophagy: The goal is to achieve recurrent states of deep cellular cleansing, eliminating damaged proteins and optimizing mitochondrial function for healthy aging.
- "Unbreakable" mental focus: Utilizing the prolonged elevation of ketone bodies (beta- hydroxybutyrate) as a brain super fuel that eliminates any trace of mental fog and provides superior cognitive clarity throughout the workday.
- Physiological adaptation to stress: Training the nervous system to operate effectively in states of low glucose availability, improving systemic resilience.

2. Functional nutrition and nutrient timing (Pillar N)

For the Performance profile, food is not just sustenance, but strategic fuel for performance and repair.

- Structural optimization: The goal is to prioritize very high quality proteins to maintain and repair muscle mass, which acts as the body's largest glucose storage and endocrine organ.
- Mastery of nutritional timing: Synchronizing the intake of complex carbohydrates and nutrients specifically with the post-workout window , taking advantage of the peak insulin sensitivity to direct energy to immediate muscle repair.
- Hormonal control through fuel: Using food to stabilize leptin and other hormones, ensuring the body stays in a state of "building and maintenance" rather than "alert and inflammation".

3. Construction of the " biological shield" (Pillar E)

The goal is to transform the body into a "healthy beast" through training geared towards specific strength and hypertrophy results.

- Periodization and progressive overload: Achieving higher strength levels through meticulous recording of loads and the use of compound movements ("The Big 5") that generate the maximum systemic and hormonal impact.
- Total halt to aging: Stop sarcopenia and activate Wolff's Law to strengthen bone mineral density through heavy loads, protecting the skeleton against future osteoporosis.
- Endocrine signaling: Using muscle to secrete myokines (such as irisin) that travel through the bloodstream protecting the health of the brain, liver, and pancreas.

4. Elite recovery and circadian optimization (Pillar U)

Sleep is considered the ultimate performance tool, not downtime.

- Biological synchronization: The goal is to perfectly align the circadian rhythm through exposure to morning sunlight and early training, optimizing the nighttime release of melatonin.
- Glymphatic brainwashing : Ensuring consistent periods of deep sleep (NREM Phase 3) to allow the brain to eliminate metabolic waste such as beta-amyloid protein, essential for the prevention of neurodegenerative diseases.
- Maximizing HGH: Harnessing the natural peak of growth hormone during deep sleep to multiply the results of training and fasting.

5. Sexual vitality as a performance marker (Pillar X)

The aim is to integrate sexual health as a biomarker of vitality and active hormonal balance.

- Natural hormonal optimization: Use physical activity and pelvic training to endogenously raise free testosterone and DHEA (the anti-aging steroid) levels.
- Cortisol management: Utilizing the release of oxytocin after intimacy as the master antagonist of cortisol, promoting muscle retention and reducing visceral fat accumulated from high-level work stress.
- Confidence and vital energy: Strengthening the emotional bond and body confidence, creating a psychological buffer that improves resilience to external stressors.

In summary, the goal for the NEXUS Performance user is to achieve the state of "Stretch Lifetime ": to stretch both the duration and quality of their life, ensuring that their biology operates at its maximum functional capacity through the disciplined synergy of these five pillars.

7. Intermittent fasting strategy

The intermittent fasting strategy for the NEXUS Performance profile represents Level 3 progression within the system, known as the 20-4 protocol or Warrior Style. This strategy is specifically designed for individuals who already have established discipline, train regularly, and have a high tolerance for discomfort in pursuit of clear goals of physical performance, mental clarity, and elite longevity.

The strategy is detailed below, with physiological and operational depth:

1. The operating protocol: model 20-4

This scheme consists of maintaining a strict fasting state for 20 hours a day, concentrating all caloric intake in a 4-hour window.

- Usual distribution: Generally, the feeding window is located at the end of the day (for example, from 6:00 p.m. to 10:00 p.m.), allowing the individual to operate throughout the workday and training in a state of deep fasting.
- Nutritional density: Due to the short window, it is imperative that meals have very high nutritional density to avoid deficiencies, prioritizing quality proteins and healthy fats.

2. Advanced physiological fundamentals

The Performance profile seeks to maximize the body's biological mechanisms:

- Maximizing autophagy: While shorter protocols initiate cellular recycling, it is in the 16-24 hour range where autophagy (internal cleaning of damaged proteins and old organelles) is powerfully activated through the AMPK protein.
- Metabolic Switch Mastery: By 20 hours, liver glycogen depletion is almost complete.6 The body is forced to use ketone bodies (beta- hydroxybutyrate) as its primary energy source. For the brain, this is a " super fuel " that provides more energy per unit of oxygen than glucose, completely eliminating brain fog and allowing for unwavering focus.
- Growth hormone (HGH) spikes: Prolonged fasting dramatically raises HGH levels (between 300% and 500%), which acts as a protective agent for muscle tissue and a powerful cell regenerator.

3. Implementation and golden rules during fasting

For this profile, execution must be technical and error-free to maintain efficiency:

- Permitted liquids: Water only, black coffee (without milk or sugar), green/black tea, and herbal infusions. Adding a pinch of sea salt to the water is recommended to maintain electrolyte balance and prevent headaches or weakness due to sodium deficiency.
- Strict prohibitions: Any calorie, even a "splash" of milk, interrupts autophagy. "Zero" drinks with artificial sweeteners should be avoided, as they can trigger an insulin response in the brain.
- Adrenaline management: Fasting naturally increases adrenaline and noradrenaline levels. Performance users should channel this energy surge to maximize their productivity on complex cognitive tasks during the morning and afternoon.

4. Synchronization with training (Pillar E)

The synergy between 20-4 fasting and exercise is one of the keys to this profile:

- Fasted training: It is recommended to perform heavy strength training towards the end of the fasting window (around hour 18 or 19). This takes advantage of peak HGH levels to protect muscle and maximize visceral fat burning.
- The refueling window: The first meal after training (breaking the fast) is critical. This is the time to introduce complex carbohydrates (tubers, fruit) along with protein, taking advantage of the peak insulin sensitivity to shuttle (transport) nutrients directly to muscle cells for repair.

5. Strategic and Gender Considerations

NEXUS warns that, even in high-performance profiles, biology rules:

- Women with high performance status are more sensitive to metabolic stress due to the kisspeptin molecule. It is recommended to adjust the protocol to a 14-10 ratio during the luteal phase (the week before menstruation) to protect progesterone balance and prevent amenorrhea.
- The 90/10 rule: This profile's commitment is 90%. This means that 90% of their decisions must be biologically correct, allowing 10% social flexibility to avoid cortisol-related issues stemming from isolation or obsessive-compulsive disorder (orthorexia).

In summary, the 20-4 fast for the Performance profile is a deep cellular maintenance and neurobiological optimization tool that allows the individual to operate at their maximum functional capacity while "stretching" their healthy life.

8. Real fooding strategy

The real fooding strategy for the NEXUS Performance profile transcends simply choosing healthy foods, becoming a highly precise, functional diet. For this profile, comprised of individuals with established discipline and a high training frequency (4 to 6 days per week), food is not just sustenance, but strategic fuel and the building blocks necessary for biological repair and elite longevity.

The following section details this strategy using the technical and operational depth that defines the method:

1. The philosophical framework: functional nutrition

For the Performance user, the goal is to optimize their biological "software" by consuming nutrients that respect their evolutionary design. The fundamental premise is to consume foods that maintain their original matrix (plants, animals, and fungi), avoiding products designed in laboratories to suppress satiety signals.

2. Systemic classification (NOVA system)

The Performance profile operates under a technical decision-making structure. The NOVA system is used to prioritize nutrient density.

- Absolute priority (90%): Group 1 foods (unprocessed or minimally processed). This includes meats, fish, eggs, fruits, vegetables, nuts, and seeds.
- Culinary ingredients (moderate use): High-quality fats such as extra virgin olive oil (EVOO), butter, ghee , or coconut oil for cooking and dressing.
- Radical elimination (<10%): Ultra-processed foods from Group 4 (soft drinks, pastries, ready-made meals) and the so-called "Trinity of Toxicity" should be avoided: liquid sugar, industrial seed oils (sunflower, corn, soy) and refined flours.

3. The three pillars of the Performance strategy

Unlike other profiles, the Performance profile focuses on three critical aspects:

A. Prioritizing high-quality protein

Protein is the most satiating macronutrient and is essential for maintaining muscle, understood as the body's largest endocrine organ. Performance users must ensure sufficient protein intake (eggs, lean meats, fish, legumes) to repair muscle damage caused by strength training and prevent catabolism during the 20-4 fasting windows.

B. Mastery in nutritional timing

This is the key differentiator of the Performance profile. The most critical time for consuming real food is the post-workout window .

- Insulin Sensitivity: After heavy strength exercise, insulin sensitivity is at its maximum.
- Glycogen replenishment: This is the ideal time to introduce complex carbohydrates (potatoes, sweet potatoes, fruit, rice) along with protein. This ensures that nutrients are directed straight to muscle cells for repair and not for fat storage.

C. Adjustment according to the stimulus (flexible fuel)

The diet should vary according to the demands of training. On days of intense training, the consumption of tubers and fruits is increased; on days of active rest, healthy fats (avocado, nuts, extra virgin olive oil) and vegetables are prioritized to maintain metabolic flexibility.

4. Implementation protocol: The 90/10 rule

Due to their ambitious performance and longevity goals, this profile elevates nutritional commitment to the 90/10 rule. 90% of decisions must be biologically correct, leaving only 10% for social flexibility, understanding that the stress (cortisol) resulting from excessive obsession (orthorexia) can be as harmful as sugar itself.

5. Practical Implementation Guide

- Perimeter hunting: The Performance user shops almost exclusively on the outer perimeter of the supermarket, where the basic fresh and refrigerated foods reside.
- 3-step label filter: We don't look at calories, we look at ingredients. We discard any product with more than 5 ingredients, with unpronounceable names, or where sugar or its aliases (maltodextrin, dextrose) are among the first three ingredients.
- Satiety management: The awareness is trained to stop eating at a satiety level of 8 out of 10, always avoiding the state of feeling "full", which indicates a lack of connection with the hormonal signals of leptin.
- Hydration with electrolytes: Drink water (half your weight in ounces) adding a pinch of Himalayan or sea salt to ensure optimal cellular absorption and maintain energy levels during prolonged fasting.

In summary, the real fooding strategy for the NEXUS Performance profile is an operating system where food functions as a high-fidelity chemical signal that communicates health and performance to every cell, ensuring that the individual not only survives, but thrives to its maximum capacity.

9. Sleep hygiene strategy

The sleep hygiene strategy for the NEXUS Performance profile transcends basic recovery to become a precision circadian optimization. For this individual, who tolerates discomfort and seeks maximum biological potential, sleep is the cornerstone that determines their hormonal balance, immune function, and cognitive acuity. At this level, sleep is not a luxury, but a biological imperative and the most powerful performance enhancer available.

The following details the comprehensive strategy for the Performance profile:

1. The core of the strategy: circadian optimization

Unlike other profiles, the Performance user uses light and *timing* as biological synchronization tools through the suprachiasmatic nucleus (SCN).

- Morning anchoring: Exposure to bright natural light (10–15 min) between 06:00 and 08:30 is required to trigger the cortisol peak necessary for alertness and stop melatonin production.
- Training timing: The Performance profile trains regularly (4–6 days per week); the ideal time for heavy strength training is between 5:00 p.m. and 7:00 p.m., when motor coordination and strength are at their circadian peak.
- Blue light blocking: At least one hour before bedtime, screen use should be eliminated, as the brain interprets blue light as midday sun, blocking the release of melatonin by the pineal gland.

2. The "10-3-2-1-0" operating protocol

This chronological framework eliminates decision fatigue and ensures entry into deep sleep phases:

- 10 hours before bed (zero caffeine): Caffeine has a half-life of 5–7 hours and blocks adenosine receptors in the brain, preventing the onset of deep sleep.
- 3 hours before bedtime (no large meals or alcohol): Digestion raises core body temperature, making it harder to fall asleep. Although alcohol is a sedative, it disrupts the second half of the night and suppresses REM sleep.
- 2 hours before sleeping (zero work stress): Emails or news that trigger cortisol should be avoided.
- 1 hour before bed (zero screens): Use blue light blocking glasses or read physical books with warm light.
- 0 (zero snooze): Do not press the alarm snooze button to avoid sleep inertia or morning grogginess.

3. Physiology of elite recovery (the "Why")

The Performance profile aims to optimize specific cellular processes during the night:

- Glymphatic system : During deep sleep (NREM Stage 3), the brain activates its "washout cycle", eliminating metabolic waste such as beta-amyloid protein.
- HGH spikes: Most growth hormone (HGH) is released during deep sleep, being vital for muscle repair after heavy training.
- Fat control: Sleep regulates leptin (satiety) and ghrelin (hunger). Sleeping less than 7 hours increases ghrelin, prompting the body to store visceral fat and crave carbohydrates.

4. The high-performance "sleep sanctuary"

allostatic load :

- Cool temperature: The ideal range is 18°C (65°F); the body needs its internal temperature to drop in order to begin resting.
- Absolute darkness: Use blackout curtains or high-quality eye masks to avoid suppressing melatonin.
- Tactical silence: Use of white noise machines to mask disruptive sounds.

5. Synergies within the NEXUS system

The sleep pillar acts as a multiplier for the results obtained in the other four:

- With strength training (E): Heavy exercise early in the day improves the depth of later sleep, but vigorous training just before bedtime can delay the onset of sleep .
- With sexual vitality (X): The release of prolactin post-orgasm is one of the most natural ways to induce the deep relaxation needed to enter NREM phase 3.
- With fasting (S) and real food (N): Eating dinner early, thanks to the 20/4 fasting method, releases energy that is redirected from digestion to cellular repair. Consuming magnesium (green vegetables) and tryptophan (eggs, turkey) during the eating window facilitates melatonin production.

In summary, for the NEXUS Performance user, mastering sleep hygiene and circadian synchronization is the ultimate biohacking strategy to ensure that every training session and every hour of work translates into active longevity and biological superiority.

10. Strength training strategy

The strength training strategy for the NEXUS Performance profile represents the most advanced and technical level of biological stimulation within the method. This profile is characterized by high-frequency training (4 to 6 days per week), a high tolerance for discomfort, and the pursuit of clear goals for physical and mental optimization, as well as elite longevity.

Under the "E" pillar (Exercise), training is not perceived as an obligation, but as the essential stimulus to keep the body young, with muscle acting as the body's largest endocrine organ.

The comprehensive strategy for this profile is detailed below:

1. The physiological framework: Muscle as biological shielding

For the Performance user, strength training is the ultimate tool for building "armor" against aging.

- Myokine signaling : When muscle contracts against heavy resistance, myokines (such as IL-6 and Irisin) are secreted, which travel through the bloodstream communicating with the brain, liver, bones, and pancreas to optimize systemic health.
- Metabolic reserve: Muscle is the body's largest glucose storage site; increased muscle mass dramatically improves insulin sensitivity and reduces the risk of metabolic diseases.
- Wolff's Law and bone density: Heavy loads generate piezoelectric signals in bones, activating osteoblasts to strengthen the bone matrix and shield the skeleton against future osteoporosis.

2. The mechanisms of growth and adaptation

The Performance profile must master the three mechanisms that trigger hypertrophy and functional strength:

1. Mechanical tension (king): This is the physical force applied to the muscle fibers. It is achieved through heavy loads, ideally above 80% of the one-repetition maximum (1RM) to recruit "high-threshold" motor units.

2. Metabolic stress: The "burning" or pumping generated by the accumulation of metabolites (lactate, hydrogen ions) that signal the cell to adapt and grow.

3. Muscle damage: Micro-tears in the fibers (especially during the eccentric or lowering phase) that trigger a reparative inflammatory response.

3. Programming: goal-oriented training

Unlike other profiles, the Performance user uses periodization and load logging to avoid stagnation.

Selection of moves: "The Big 5"

The training is based on compound movements that maximize hormonal impact and coordination:

- Squat : Knee dominant; uses the largest muscle groups.
- Deadlift : The purest test of strength; builds a "shielded" back.
- Bench Press: Horizontal pushing force of the upper body.
- Military press (Overhead Press): Vertical push; requires extreme core stability .
- Pull-ups or barbell rows: Vertical and horizontal pulling; essential for posture.

Operational Variables

- Frequency: 4 to 6 days per week with detailed planning.
- Repetition ranges: It is recommended to spend 70% of the time in the 5-8 repetition range, seeking the optimal balance between neurological strength and muscle growth.
- Progressive overload: It is the immutable law; the body only changes if the stimulus increases (adding weight, volume, density or improving technique).
- Rest intervals: To lift heavy effectively, 2 to 3 minutes of rest are required between sets to allow cellular ATP to be replenished.

4. Impact on longevity and cardiovascular health

The science behind NEXUS shows that strength training is a "life insurance":

- Mortality reduction: Any amount of strength training reduces the risk of mortality from all causes by between 15% and 21%.
- Synergistic effect : The greatest benefit for longevity is obtained by combining strength with aerobic activity (MVPA), achieving a reduction in the risk of mortality of up to 40%.
- Blood pressure control: Dynamic training of moderate intensity generates significant reductions in systolic and diastolic blood pressure.

5. Synergy with other NEXUS pillars

The exercise pillar works in sync with the others to maximize elite recovery:

- With fasting (S): Weight training towards the end of the fasting window raises growth hormone (HGH) by 300% to 500%, protecting muscle and optimizing regeneration.
- With nutrition (N): The post-workout window is the most critical time to ingest real carbohydrates and protein, taking advantage of the peak insulin sensitivity to repair damaged tissue.
- With sleep (U): Muscle damage is repaired during deep sleep (NREM stage 3); training hard early in the day improves the depth of later rest.
- With sexual vitality (X): Heavy compound exercises are the most effective natural way to raise free testosterone, which is the main driver of libido and vital energy.

In summary, the Performance strategy for strength training does not only seek aesthetics, but to turn the individual into a "healthy beast" through a high-frequency and precision system that stretches *functional lifetime* .

11. Sexual life strategy

The active sex life strategy for the NEXUS Performance profile constitutes the fifth pillar of the method, identified with the letter "X" (eXpression). For this individual, with an established discipline and a high tolerance for discomfort, sexuality is not approached solely as a recreational activity, but as a tool for biological optimization, elite performance, and longevity.

The specific strategy for this profile is called "Sexuality as performance" and is detailed below in all its technical and scientific depth:

1. The rationale: Sexual health as an elite biomarker

At the Performance profile level, a fulfilling sex life is considered a critical biomarker that reflects the success of the other pillars (nutrition, strength, and rest). This activity acts as a neurobiological regulatory force that supports mood stability and hormonal balance, functioning as a necessary counterweight to the chronic stress of a high-performance lifestyle.

2. Optimization of the "hormonal cascade"

The Performance user uses sexual activity to positively manipulate their internal chemical environment:

- Cortisol antagonism (Oxytocin): Physical contact and orgasm release oxytocin, which directly counteracts the effects of cortisol. For someone who trains hard, this translates to better muscle mass retention and a reduction in stress-induced abdominal fat storage.
- DHEA (The anti-aging steroid): Sexuality temporarily raises levels of DHEA, a precursor to testosterone and estrogen, which strengthens immune function and improves bone density.
- Dopamine peak and focus: The anticipation and execution of the sexual act increase dopamine, which improves daily motivation and acts as a buffer against depressive symptoms.
- Induction of deep sleep (Prolactin): After orgasm, the release of prolactin induces a state of natural relaxation that facilitates entry into NREM Phase 3 of deep sleep, multiplying nighttime cellular recovery.

Structural strengthening and prevention

Beyond the hormonal aspect, this pillar offers direct physical benefits for long-term performance and health:

- Cardiovascular and endothelial health: Sex is a form of moderate exercise that improves the function of the endothelium (the lining of blood vessels), promoting systemic blood flow and reducing the risk of cardiovascular disease.
- Prostate health (men): Frequent ejaculation is part of the body's natural "cleaning" process, associated with a reduced risk of prostate cancer in later stages of life.
- Pelvic and core training (women): Strengthening the pelvic floor muscles is synergistic with heavy strength work of the E-pillar, improving core stability and bladder control.
- Pain resilience: Endorphins released during orgasm act as powerful natural painkillers, which can improve recovery after intense workouts.

4. Operational tactics for the Performance profile

To prevent the pace of life from sacrificing this pillar, the method proposes:

- Strategic scheduling: Similar to training, scheduling intimate moments ensures that the activity remains a sustained health priority, eliminating the stress of uncertainty.
- Quality over frequency: The focus is on deep connection and reducing performance anxiety, thus maximizing psychological and hormonal benefits.
- Body confidence: Maintaining a good physical composition through the method reinforces self-image, which is key to a healthy vital expression.

5. Systemic synergy with the NEXUS pillars

The success of the X pillar for the Performance user depends on full integration:

- With training (E): Heavy compound exercises (squats, deadlifts) are the most effective way to increase free testosterone, which is the main driver of libido and vital energy.
- With nutrition (N): Consuming "real food" provides essential nutrients such as Zinc, Selenium and Vitamin D, indispensable building blocks for the production of sex hormones.
- With sleep (U): Poor sleep drastically reduces testosterone; prioritizing rest is the "fuel" needed to maintain intimate vitality.

In conclusion, for the Performance profile, sexual vitality is the ultimate expression of optimized biology, functioning as a master regulator that protects muscle, heart, and mind while "stretching" *healthy lifespan* .

12. Conclusions and action plan

For the NEXUS Performance profile, the method represents the pinnacle of conscious biological optimization, designed for individuals who seek not only health, but elite functional longevity. This level of commitment requires transforming every habit into a precision tool for stretching healthy life *through the* synergy of high-intensity stimuli and deep recovery protocols.

12.1. Conclusions: The biological impact on the performance profile

The integration of the pillars in the Performance profile generates a superior physiological state based on the following fundamentals:

1. Metabolic Switch Mastery and Autophagy: By implementing a 20/4 fast (Warrior Style), the body almost completely depletes its glycogen stores, forcing the brain to use ketone bodies (beta- hydroxybutyrate) as a " super fuel " that clears brain fog and enhances cognitive sharpness. This prolonged period powerfully activates autophagy, the cellular recycling mechanism that eliminates damaged proteins and old organelles, acting as a shield against neurodegenerative diseases.

2. Muscle as an endocrine organ and life insurance: At this level, strength training is not aesthetic, but functional; the muscle is trained to secrete myokines (such as IL-6 and irisin) that regulate the metabolism of the liver and pancreas and protect brain health. The use of heavy loads (>80% 1RM) is the only way to recruit "high-threshold" motor fibers, which prevent sarcopenia and activate Wolff's Law, strengthening bone density against osteoporosis.

3. Circadian synchronization and brain cleansing: Sleep is approached as a biological imperative where the glymphatic system performs a nightly "washing" of metabolic waste products such as beta-amyloid protein. Circadian optimization, through exposure to morning sunlight and training at peak evening strength, ensures that the majority of growth hormone (HGH) is released during deep sleep, amplifying the cellular repair initiated by fasting.

4. Sexual vitality as an elite biomarker: A fulfilling sex life acts as a master stress regulator; the release of oxytocin during orgasm functions as a direct antagonist of cortisol, promoting muscle mass retention and reducing abdominal fat induced by work-related stress. Furthermore, it temporarily raises levels of DHEA, the anti-aging steroid that supports immune function and bone density.

12.2. Plan (print this sheet)

This plan is designed to maximize physical and mental performance through a highly disciplined structure and minimal operational friction.

1. Pilar S (Stress Adaptation): Strict 20-4 Protocol
    - Fasting window: 20 hours of strict fasting per day (e.g., from 10:00 PM to 6:00 PM the following day).
    - Technical hydration: During fasting, consume at least 2-3 liters of water with a pinch of sea salt to maintain electrolyte balance and blood pressure.
    - Adrenaline management: Use the natural peak of noradrenaline and adrenaline generated by fasting to perform the most cognitively complex tasks during the morning.
2. Pilar N (Proper Nutrition): Functional Nutrition 90/10
    - Golden rule: 90% of intake should be real food that maintains its original matrix (plants, animals, fungi), radically eliminating the "Trinity of Toxicity" (liquid sugar, industrial seed oils and refined flours).
    - Protein prioritization: Ensure high nutrient density within the 4-hour window, prioritizing high-quality protein for muscle repair.
    - Nutritional timing: Consume most complex carbohydrates (potato, fruit, rice) in the post-strength training meal to take advantage of maximum insulin sensitivity and direct nutrients to muscle tissue.
3. Pillar E (Exercise): Goal-oriented training
    - Frequency: 4 to 6 days per week with meticulous recording of loads in a logbook .
    - Compound movements (The Big 5): Focus the routine on Squat, Deadlift, Bench Press, Military Press and Pull-ups/Rowing to maximize systemic hormonal impact.
    - Progressive overload: Try adding between 1.25kg and 2.5kg to the bar in each session or increase training density (same work in less time).
    - Strategic rest: Maintain 2-3 minute intervals between heavy sets to allow for full cellular ATP resynthesis.

4. Pilar U (Unwind): Elite Circadian Optimization
    - Morning anchoring: Get 10-15 minutes of direct sunlight before 8:30 a.m. to synchronize your circadian rhythm and suppress residual melatonin.
    - Protocol 10-3-2-1-0:
        - 10 hours before: Zero caffeine .
        - 3 hours before: No large meals (the 20-4 fast makes this easier by eating dinner early).
        - 2 hours before: Zero work or digital stress.
        - 1 hour before: Zero screens (total blue light blocking).
        - 0: Get up immediately when the alarm sounds (without snooze) to avoid the inertia of sleep.
    - The sanctuary: Keep the bedroom at 18°C, in absolute darkness and total silence.
5. Pillar X (eXpression): Sexuality as performance
    - Intentionality: Schedule moments of intimacy to ensure that this hormonal regulator is a priority in a high-performance schedule.
    - Focus on connection: Prioritize quality and reducing performance anxiety to maximize prolactin release, which facilitates entry into NREM stage 3 deep sleep.
    - Synergy with pillar E: Use heavy strength training to naturally raise free testosterone, the main driver of vital energy and libido.

Weekly Performance Profile Execution Summary :

Day	Pilar S (20-4)	Pilar N (Food)	Pillar E (Strength)	Pilar U (Dream)
Monday	20h Fasting	Functional	Force (Objective A)	Circadian Optimization
Tuesday	20h Fasting	Functional	Strength (Objective B)	Circadian Optimization
Wednesday	20h Fasting	Functional	Active rest / Mobility	Circadian Optimization
Thursday	20h Fasting	Functional	Force (Objective C)	Circadian Optimization
Friday	20h Fasting	90/10 (Social)	Force (Objective A)	Circadian Optimization
Saturday	16h Flexible	90/10 (Social)	Specific training	Focus on Pillar X
Sunday	Circadian	Functional	Full recovery	Full recovery

Discipline is not a burden, but the foundation of your biological freedom. Consistency in these five pillars will create a biological armor that will not only allow you to perform today, but to thrive for decades to come.

13. Final message for the user

You've taken the first step toward reclaiming sovereignty over your own biology, and that in itself is a victory. The NEXUS method isn't a fad diet or a set of tricks; it's a coherent life framework designed to align you with your evolutionary blueprint. You're about to hit your body's "reset button" to clear system errors and optimize your biological "software."

However, for this journey to be successful, you must arm yourself with the most powerful tool for longevity: patience. We live in a culture of instant gratification, but your physiology doesn't operate at the click of a button. Your body needs between two and four weeks to become metabolically flexible and learn to burn fat efficiently after years of glucose dependence. You're not looking for a superficial aesthetic change for the next month; you're building armor for the rest of your life.

Remember that the five pillars work in perfect synergy:

- Real Nutrition (N) provides the building blocks without hijacking your brain with industrial additives.
- Strength training (E) turns your muscle into an endocrine organ that communicates health to your entire body.
- Sexual Vitality (X) is your master stress regulator, restoring your hormonal balance.
- Sleep (U) is the "brainwashing" that repairs daily damage.
- Fasting (S) activates your "internal pharmacy" and the cellular recycling of autophagy.

Be aware that the biggest mistake is falling into the trap of perfectionism. NEXUS is based on the principle that consistency beats perfection. That's why we apply the 80/20 or 90/10 rule: if the vast majority of your decisions are biologically sound, your body will have the resilience needed to handle social flexibility without guilt or setbacks.

Don't compare yourself to someone else's profile; this method is an "operating system" that adapts to your stage of life, whatever it may be. If you slip up one day, don't punish yourself: simply return to the pattern the next day. Your ultimate goal is "Stretch Lifetime ": stretching not only the length of your life, but the quality of every minute you spend in it, ensuring you reach your golden years with energy, mental clarity, and independence.

Trust the process, listen to your body, and keep going. You're designed to thrive, not just survive. Welcome to your best self!

14. NEXUS Support

This is a practical guide designed to help you integrate the five core pillars of the NEXUS Method into your daily life in a simple and effective way. Its purpose is to provide clarity and a strong foundation to build your health, energy, and longevity. However, if you are looking for a deeper and more personalized approach, you can access NEXUS Support, our advanced coaching service. Discover how to unlock the full potential of the method on our website.